# HL7 Interfacing and RODS

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# HL7 Interfacing

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- Health Level Seven Overview

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- HL7 Tools
  - Integration Engine (MIRTH)
  - Simulators (nule.org)

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- HL7 Tools
  - Integration Engine (MIRTH)
  - Simulators (nule.org)
- Tasks
  - HL7 Routing
  - Filtering
  - Transformation

# HL7 Overview

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- Source for HL7 tutorial information:
  http://www.interfaceware.com/manual/hl7.html

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- Standard (HL7.org)

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- HL7 Version 3.0 in development since 1997 (XML Schema, RIM)

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- Disadvantage
  - Not plug and play – semantic, vocabulary differences in implementation by different vendors

# Message Structure

- Message
  - Segments - MSH, PID, NK1, PV1
    - Fields - |CONROY^MARI^^^^|
      - Subfields - CONROY

```
MSH|^~\&|EPIC|EPICADT|SMS|SMSADT|199912271408|CHARRIS|ADT^A04|1817457|D|2.3|
PID||0493575^^^2^ID 1|454721||DOE^JOHN^^^^|DOE^JOHN^^^^|19480203|M||B|254
 E238ST^^EUCLID^OH^44123^USA||(216)731-4359|||M|NON|400003403~1129086|
NK1||CONROY^MARI^^^^|SPO||(216)731-4359||EC||||||||||||||||||||||||||
PV1||O|168   ~219~C~PMA^^^^^^^^^^||||277^ALLEN FADZL^BONNIE^^^^|||||||||||
 2688684||||||||||||||||||||||||199912271408||||||002376853
```

# Example Segments

- MSH segment contains information about the Sender and Receiver
- PID contains demographic information
- NK1 contains demographic information about next of kin
- PV1 contains information regarding the patient's stay in the hospital
- Z contains anything a hospital wants
- Many, many more segments in the specification

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- Found in Section 2.16 and 2.A of HL7 2.5 Specification

# HL7 Delimiters

| Character | Purpose |
|---|---|
| 0x0D | Marks the end of each segment |
| \| | Field Delimiter |
| ^ | Sub Field Delimiter |
| & | Sub-sub Field Delimiter |

'~' repetition character

'/' escape character

# Reading the Specification

- Each message is an event that has:
  - Message level definition
    - Required Segments
    - [Optional Segments]
    - {Repeating Segments}
  - Segment level definition
    - Fields - optionality, data type, repeatability

| ADT^A04^ADT A01 | ADT Message | Status | Chapter |
|---|---|---|---|
| MSH | Message Header | | 2 |
| [ { SFT } ] | Software Segment | | 2 |
| EVN | Event Type | | 3 |
| PID | Patient Identification | | 3 |
| [ PD1 ] | Additional Demographics | | 3 |
| [{ ROL }] | Role | | 15 |
| [{ NK1 }] | Next of Kin / Associated Parties | | 3 |
| PV1 | Patient Visit | | 3 |
| [ PV2 ] | Patient Visit - Additional Info. | | 3 |
| [{ ROL }] | Role | | 15 |
| [{ DB1 }] | Disability Information | | 3 |
| [{ OBX }] | Observation/Result | | 7 |
| [{ AL1 }] | Allergy Information | | 3 |
| [{ DG1 }] | Diagnosis Information | | 6 |
| [ DRG ] | Diagnosis Related Group | | 6 |
| [{ | --- PROCEDURE begin | | |
| PR1 | Procedures | | 6 |

HL7 Attribute Table - PID - Patient Identification

| SEQ | LEN | DT | OPT | RP/# | TBL# | ITEM# | ELEMENT NAME |
|---|---|---|---|---|---|---|---|
| 1 | 4 | SI | O | | | 00104 | Set ID - PID |
| 2 | 20 | CX | B | | | 00105 | Patient ID |
| 3 | 250 | CX | R | Y | | 00106 | Patient Identifier List |
| 4 | 20 | CX | B | Y | | 00107 | Alternate Patient ID - PID |
| 5 | 250 | XPN | R | Y | | 00108 | Patient Name |
| 6 | 250 | XPN | O | Y | | 00109 | Mother’s Maiden Name |
| 7 | 26 | TS | O | | | 00110 | Date/Time of Birth |
| 8 | 1 | IS | O | | 0001 | 00111 | Administrative Sex |
| 9 | 250 | XPN | B | Y | | 00112 | Patient Alias |
| 10 | 250 | CE | O | Y | 0005 | 00113 | Race |
| 11 | 250 | XAD | O | Y | | 00114 | Patient Address |
| 12 | 4 | IS | B | | 0289 | 00115 | County Code |
| 13 | 250 | XTN | O | Y | | 00116 | Phone Number - Home |
| 14 | 250 | XTN | O | Y | | 00117 | Phone Number - Business |
| 15 | 250 | CE | O | | 0296 | 00118 | Primary Language |
| 16 | 250 | CE | O | | 0002 | 00119 | Marital Status |
| 17 | 250 | CE | O | | 0006 | 00120 | Religion |
| 18 | 250 | CX | O | | | 00121 | Patient Account Number |
| 19 | 16 | ST | B | | | 00122 | SSN Number - Patient |
| 20 | 25 | DLN | B | | | 00123 | Driver's License Number - Patient |
| 21 | 250 | CX | O | Y | | 00124 | Mother's Identifier |
| 22 | 250 | CE | O | Y | 0189 | 00125 | Ethnic Group |
| 23 | 250 | ST | O | | | 00126 | Birth Place |
| 24 | 1 | ID | O | | 0136 | 00127 | Multiple Birth Indicator |
| 25 | 2 | NM | O | | | 00128 | Birth Order |
| 26 | 250 | CE | O | Y | 0171 | 00129 | Citizenship |
| 27 | 250 | CE | O | | 0172 | 00130 | Veterans Military Status |
| 28 | 250 | CE | B | | 0212 | 00739 | Nationality |

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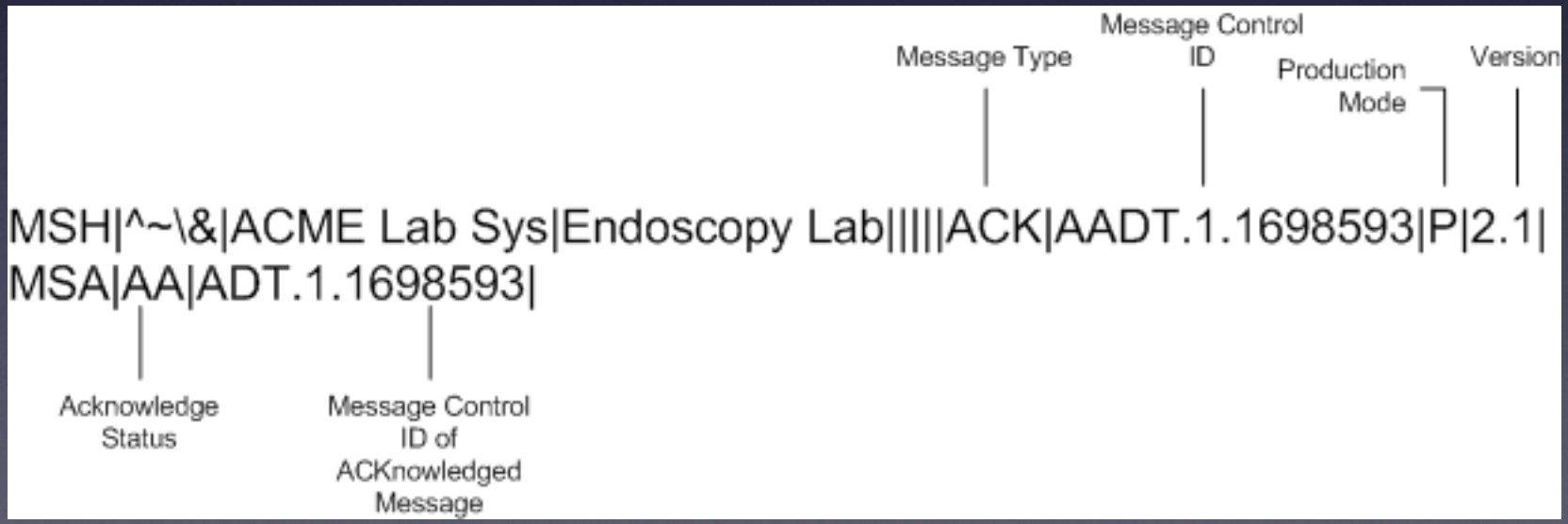

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  - MLLP uses specific ASCII characters to denote the start and end of each message
  - <SB><HL7 MESSAGE><EB><CR>
- "Batch HL7" accomplished using text files and any available file transfer protocol

# HL7 Tools

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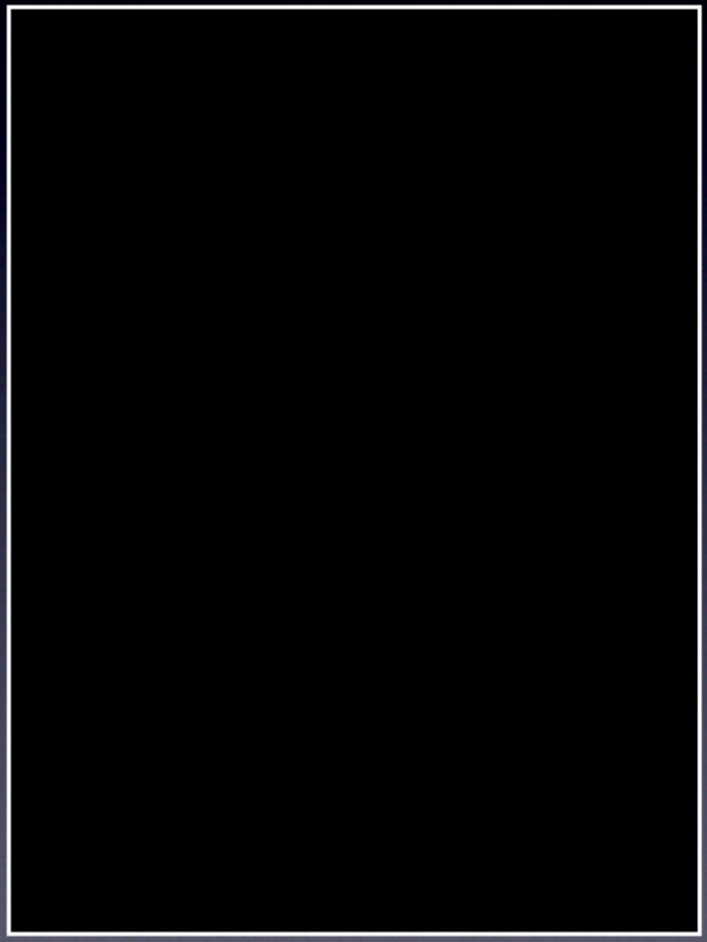
- Transform messages

# MIRTH



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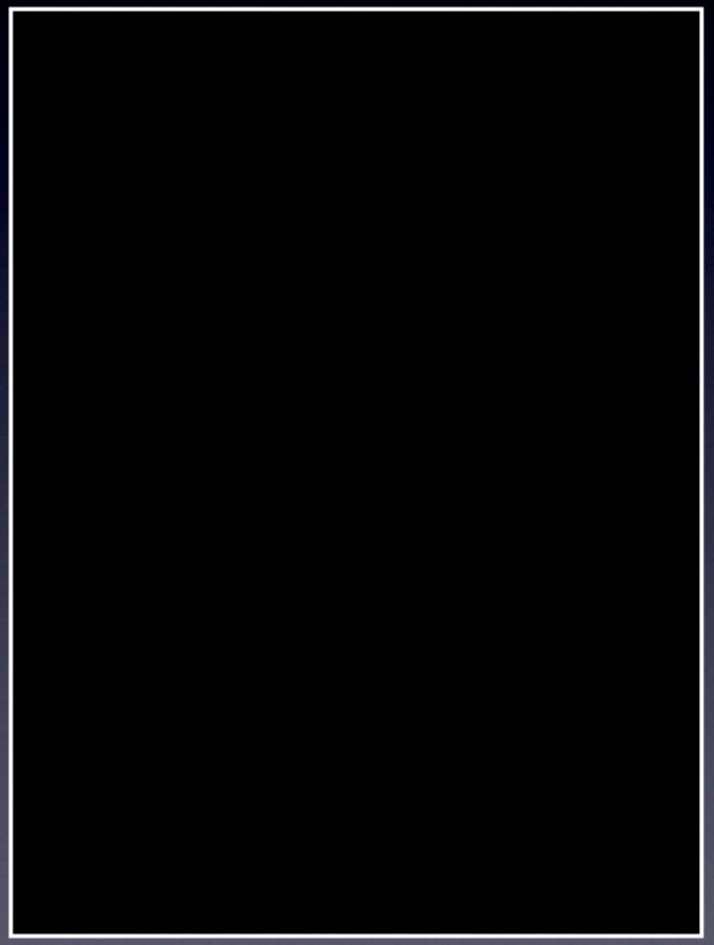

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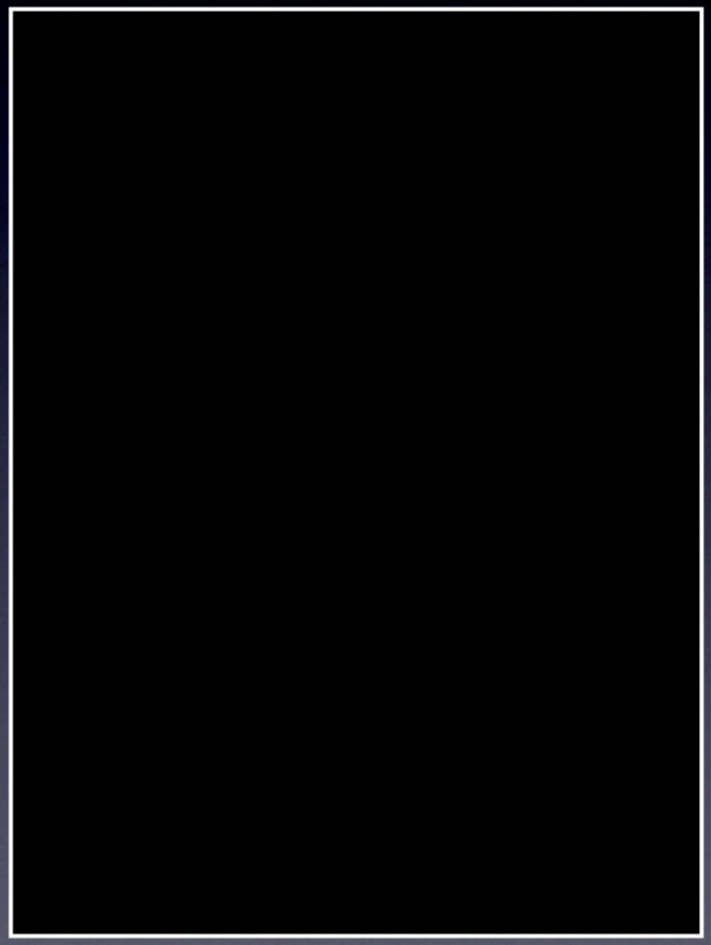

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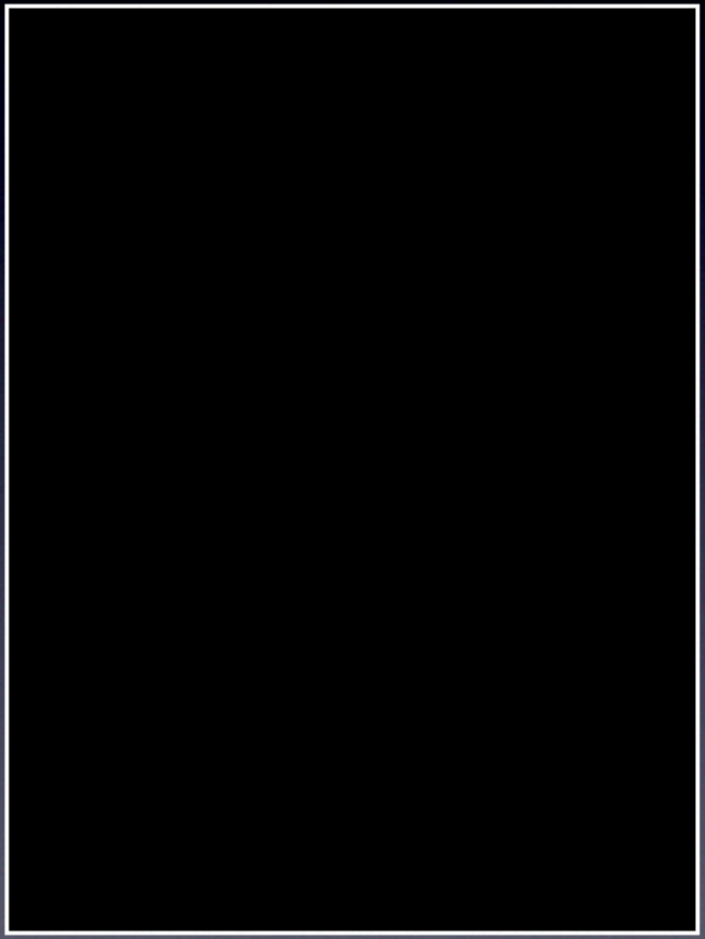

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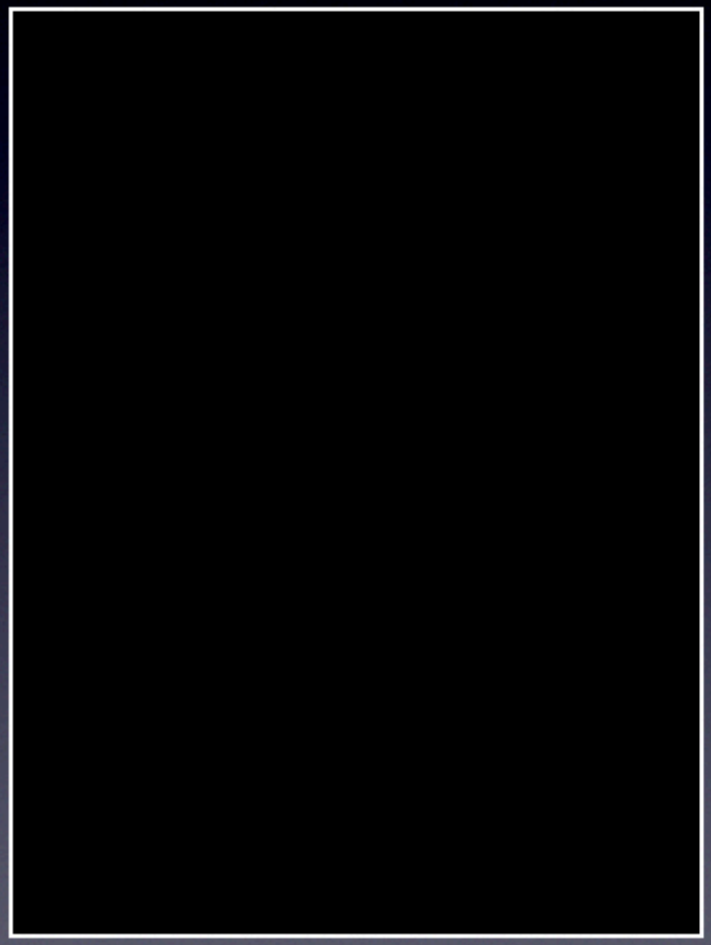
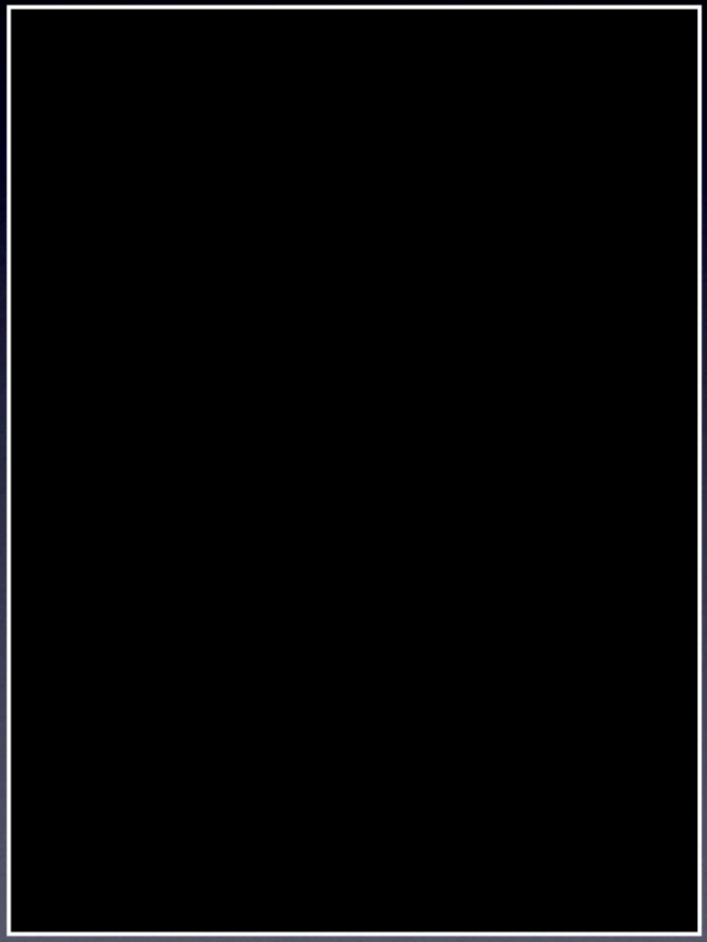
- Transformation and filtering using Javascript E4X



# Nule.org HL7 Browser

- Open source HL7 interface simulator
- An HL7 interface simulator and browser
- Inbound and outbound HL7 interfaces



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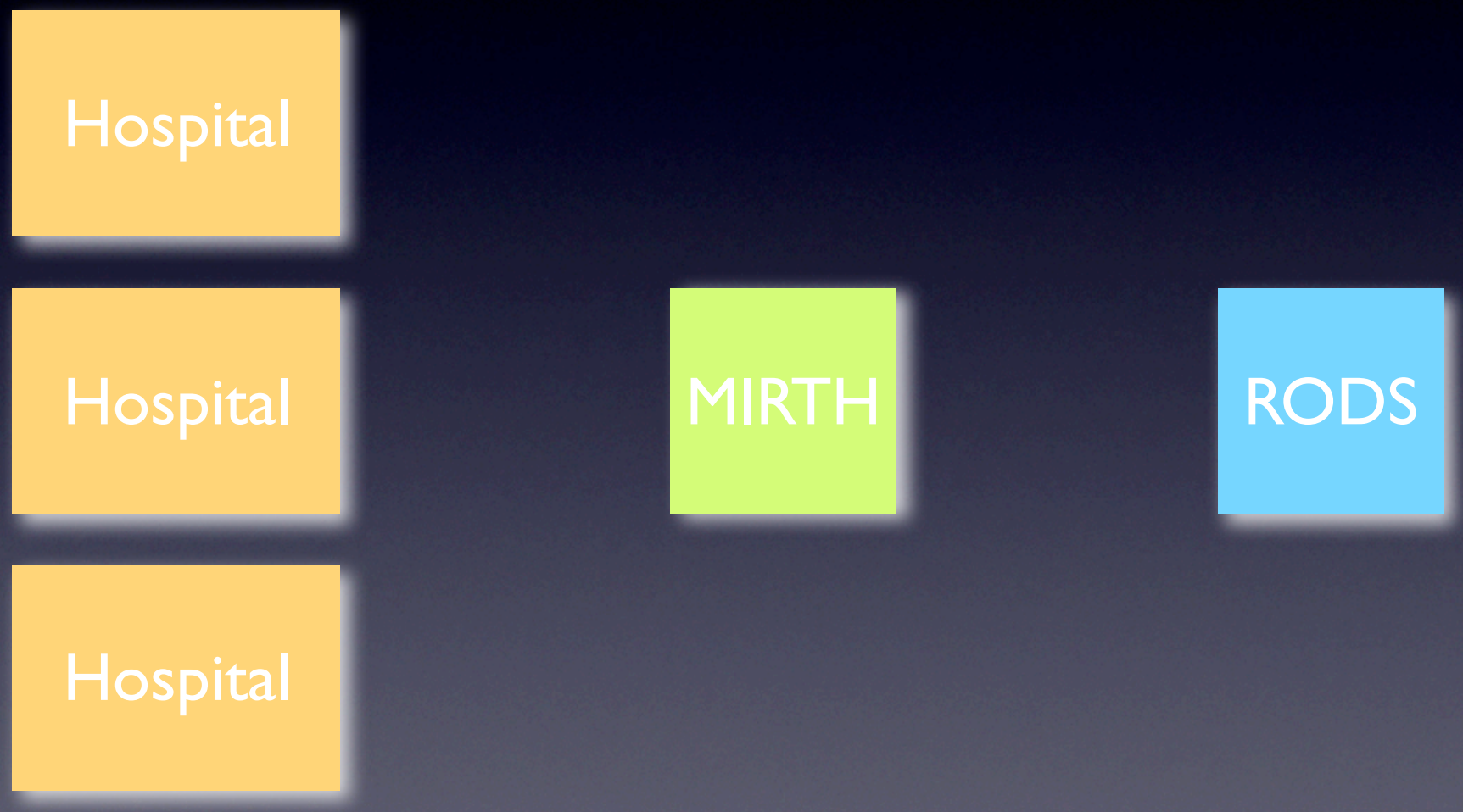

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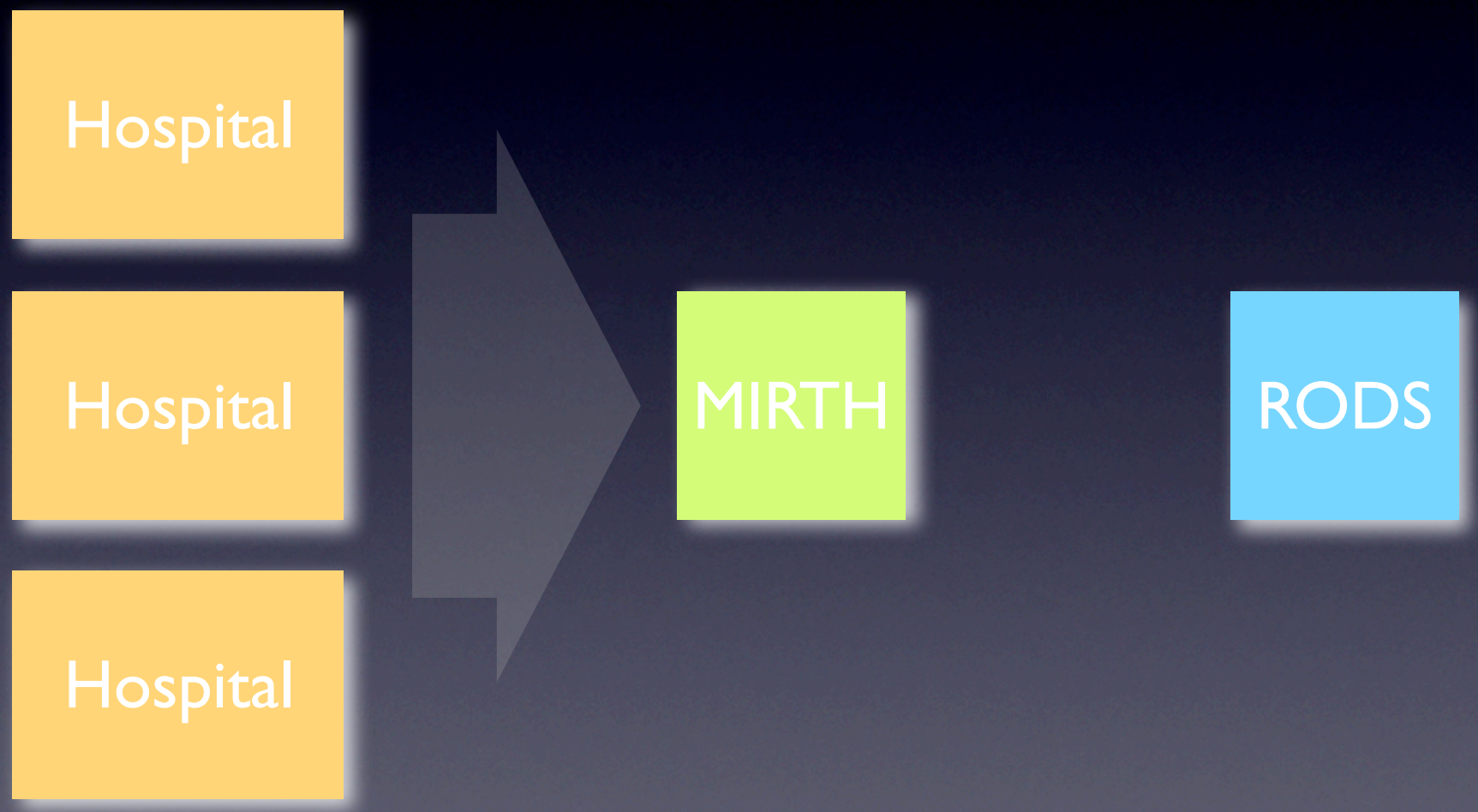

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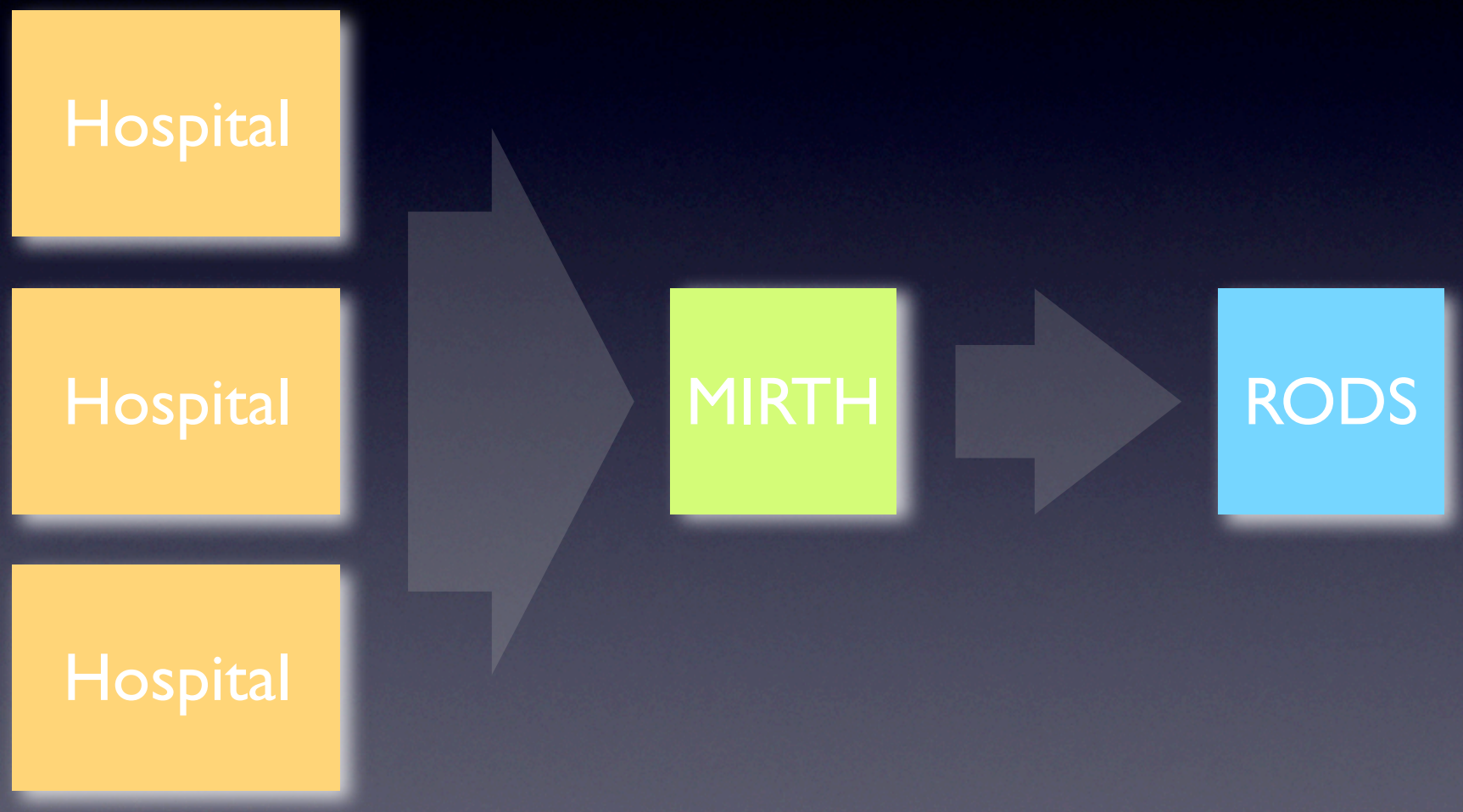

# Hands-on Lessons

# Lesson 1: Setup Simulation Environment

# Lesson 2
# Route messages

# Lesson 3
# Filter messages

# Lesson 4
# Delete fields

# Lesson 5
# Transform fields